

Netflix Data Analysis

```
In [7]: # importing lib
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

In [8]: df = pd.read_csv('nyov1adob.csv', lineterminator = '\n')

In [14]: df.head()

Out[14]:
   Release_Date  Title  Overview  Popularity  Vote_Count  Vote_Average  Original_Language  Genre  Poster_Url
0    2021-12-15  Spider-Man: No Way Home  Peter Parker is unmasked and no longer able to...  5083.954      8940         8.3          en  Action, Adventure, Science Fiction  https://image.tmdb.org/t/p/original/1g0dhYtq6L...
1    2022-03-01    The Batman  In his second year of fighting crime, Batman u...  3827.658      1151         8.1          en  Crime, Mystery, Thriller  https://image.tmdb.org/t/p/original/74xTEgt7R3...
2    2022-02-25    No Exit  Stranded at a rest stop in the mountains durin...  2618.087       122         6.3          en  Thriller  https://image.tmdb.org/t/p/original/vDHsLnOWK3...
3    2021-11-24    Encanto  The tale of an extraordinary family, the Madri...  2402.201      5076         7.7          en  Animation, Comedy, Family, Fantasy  https://image.tmdb.org/t/p/original/4jOPNkM5...
4    2021-12-22    The King's Man  As a collection of history's worst tyrants and...  1895.511      1793         7.0          en  Action, Adventure, Thriller, War  https://image.tmdb.org/t/p/original/aq4Pw5Xeu...

In [10]: df.info()
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9827 entries, 0 to 9826
Data columns (total 9 columns):
 #   Column      Non-Null Count  Dtype
---  --
 0   Release_Date  9827 non-null    object
 1   Title         9827 non-null    object
 2   Overview      9827 non-null    object
 3   Popularity    9827 non-null    float64
 4   Vote_Count   9827 non-null    int64
 5   Vote_Average  9827 non-null    float64
 6   Original_Language  9827 non-null    object
 7   Genre        9827 non-null    object
 8   Poster_Url   9827 non-null    object
dtypes: float64(2), int64(1), object(6)
memory usage: 691.1+ KB

In [11]: df['Genre'].head()

Out[11]:
0    Action, Adventure, Science Fiction
1         Crime, Mystery, Thriller
2         Crime, Mystery, Thriller
3    Animation, Comedy, Family, Fantasy
4    Action, Adventure, Thriller, War
Name: Genre, dtype: object

In [12]: df.duplicated()

Out[12]:
0    False
1    False
2    False
3    False
4    False
      ...
9822   False
9823   False
9824   False
9825   False
9826   False
Length: 9827, dtype: bool

In [15]: df.duplicated().sum()

Out[15]:
np.int64(0)

In [16]: df.describe()

Out[16]:
   Popularity  Vote_Count  Vote_Average
count  9827.000000    9827.000000    9827.000000
mean      40.326088    1392.805536     6.439534
std     108.873998    2611.206907     1.129759
min      13.354000         0.000000     0.000000
25%     16.128500    146.000000     5.900000
50%     21.199000    444.000000     6.500000
75%     35.191500   1376.000000     7.100000
max     5083.954000  31077.000000    10.000000
```

Exploration Summary

- we have a df consisting of 9827 rows and 9 columns
- our dataset look like a bit tidy with no NaNs nor duplicate values
- Release Date column needs to be casted into date time
- Overview, Original Language, PosterUrl wouldn't be so use so well deep them
- Genre column has csv and white spaces that needs to be handled and casted into category

```
In [17]: df.head()

Out[17]:
   Release_Date  Title  Overview  Popularity  Vote_Count  Vote_Average  Original_Language  Genre  Poster_Url
0    2021-12-15  Spider-Man: No Way Home  Peter Parker is unmasked and no longer able to...  5083.954      8940         8.3          en  Action, Adventure, Science Fiction  https://image.tmdb.org/t/p/original/1g0dhYtq6L...
1    2022-03-01    The Batman  In his second year of fighting crime, Batman u...  3827.658      1151         8.1          en  Crime, Mystery, Thriller  https://image.tmdb.org/t/p/original/74xTEgt7R3...
2    2022-02-25    No Exit  Stranded at a rest stop in the mountains durin...  2618.087       122         6.3          en  Thriller  https://image.tmdb.org/t/p/original/vDHsLnOWK3...
3    2021-11-24    Encanto  The tale of an extraordinary family, the Madri...  2402.201      5076         7.7          en  Animation, Comedy, Family, Fantasy  https://image.tmdb.org/t/p/original/4jOPNkM5...
4    2021-12-22    The King's Man  As a collection of history's worst tyrants and...  1895.511      1793         7.0          en  Action, Adventure, Thriller, War  https://image.tmdb.org/t/p/original/aq4Pw5Xeu...

In [18]: df['Release_Date'] = pd.to_datetime(df['Release_Date'])
print(df['Release_Date'].dtypes)
#converting release date into date time type
datetime64[ns]

In [19]: df['Release_Date'] = pd.to_datetime(df['Release_Date']).dt.year
df['Release_Date'].dtypes

Out[19]:
dtype('int32')

In [20]: df.head()

Out[20]:
   Release_Date  Title  Overview  Popularity  Vote_Count  Vote_Average  Original_Language  Genre  Poster_Url
0    2021      Spider-Man: No Way Home  Peter Parker is unmasked and no longer able to...  5083.954      8940         8.3          en  Action, Adventure, Science Fiction  https://image.tmdb.org/t/p/original/1g0dhYtq6L...
1    2022      The Batman  In his second year of fighting crime, Batman u...  3827.658      1151         8.1          en  Crime, Mystery, Thriller  https://image.tmdb.org/t/p/original/74xTEgt7R3...
2    2022      No Exit  Stranded at a rest stop in the mountains durin...  2618.087       122         6.3          en  Thriller  https://image.tmdb.org/t/p/original/vDHsLnOWK3...
3    2021      Encanto  The tale of an extraordinary family, the Madri...  2402.201      5076         7.7          en  Animation, Comedy, Family, Fantasy  https://image.tmdb.org/t/p/original/4jOPNkM5...
4    2021      The King's Man  As a collection of history's worst tyrants and...  1895.511      1793         7.0          en  Action, Adventure, Thriller, War  https://image.tmdb.org/t/p/original/aq4Pw5Xeu...

In [21]: # dropping the columns
cols = ['Overview', 'Original_Language', 'Poster_Url']

In [22]: df.drop(cols, axis = 1, inplace = True)
df.columns

Out[22]:
Index(['Release_Date', 'Title', 'Popularity', 'Vote_Count', 'Vote_Average',
      'Genre'],
      dtype='object')

In [23]: df.head(3)

Out[23]:
   Release_Date  Title  Popularity  Vote_Count  Vote_Average  Genre
0    2021  Spider-Man: No Way Home    5083.954      8940         8.3  Action, Adventure, Science Fiction
1    2022      The Batman    3827.658      1151         8.1  Crime, Mystery, Thriller
2    2022      No Exit    2618.087       122         6.3  Thriller

categorizing VoteAverage column

• We would cut the Vote_Average values and make 4 categories: popular average below_avg not_popular to describe it more using categorize_col() function provided above

In [24]: def categorize_col(df, col, labels):
    edges = [df[col].describe()[['min'],
df[col].describe()[['25%'],
df[col].describe()[['50%'],
df[col].describe()[['75%'],
df[col].describe()[['max']]

    df[col] = pd.cut(df[col], edges, labels = labels, duplicates = 'drop')
    return df

#setting the edges to cut the column accordingly

In [25]: labels = ['not_popular', 'below_avg', 'average', 'popular']
categorize_col(df, 'Vote_Average', labels)

df['Vote_Average'].unique()

Out[25]:
['popular', 'below_avg', 'average', 'not_popular', NaN]
Categories (4, object): ['not_popular' < 'below_avg' < 'average' < 'popular']

In [26]: df.head()

Out[26]:
   Release_Date  Title  Popularity  Vote_Count  Vote_Average  Genre
0    2021  Spider-Man: No Way Home    5083.954      8940      popular  Action, Adventure, Science Fiction
1    2022      The Batman    3827.658      1151      below_avg  Crime, Mystery, Thriller
2    2022      No Exit    2618.087       122      below_avg  Thriller
3    2021      Encanto    2402.201      5076      popular  Animation, Comedy, Family, Fantasy
4    2021      The King's Man    1895.511      1793      average  Action, Adventure, Thriller, War

In [27]: df['Vote_Average'].value_counts()

Out[27]:
Vote_Average
not_popular    2467
popular         269
average        282
below_avg     2398
Name: count, dtype: int64

In [28]: df.dropna(inplace = True)
df.isna().sum()

Out[28]:
Release_Date      0
Title             0
Popularity        0
Vote_Count       0
Vote_Average     0
Genre            0
dtype: int64

In [29]: df.head()

Out[29]:
   Release_Date  Title  Popularity  Vote_Count  Vote_Average  Genre
0    2021  Spider-Man: No Way Home    5083.954      8940      popular  Action, Adventure, Science Fiction
1    2022      The Batman    3827.658      1151      popular  Crime, Mystery, Thriller
2    2022      No Exit    2618.087       122      below_avg  Thriller
3    2021      Encanto    2402.201      5076      popular  Animation, Comedy, Family, Fantasy
4    2021      The King's Man    1895.511      1793      average  Action, Adventure, Thriller, War
```

- we'd split genres into a list and then explode our dataframe to have only one genre per row for each movie

```
In [32]: df = df.dropna(subset=['Genre'])
df['Genre'] = df['Genre'].str.split(',')
df = df.explode('Genre').reset_index(drop=True)

In [33]: df.head()

Out[33]:
   Release_Date  Title  Popularity  Vote_Count  Vote_Average  Genre
0    2021  Spider-Man: No Way Home    5083.954      8940      popular  Action
1    2021  Spider-Man: No Way Home    5083.954      8940      popular  Adventure
2    2021  Spider-Man: No Way Home    5083.954      8940      popular  Science Fiction
3    2022      The Batman    3827.658      1151      popular  Crime
4    2022      The Batman    3827.658      1151      popular  Mystery

In [37]: # casting column into category
df['Genre'] = df['Genre'].astype('category')
df['Genre'].dtypes

Out[37]:
CategoricalDtype(categories=['Action', 'Adventure', 'Animation', 'Comedy', 'Crime',
                             'Documentary', 'Drama', 'Family', 'Fantasy', 'History',
                             'Horror', 'Music', 'Mystery', 'Romance', 'Science Fiction',
                             'TV Movie', 'Thriller', 'War', 'Western'],
                  ordered=False, dtype=object)

In [37]: df.info()
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25552 entries, 0 to 25551
Data columns (total 6 columns):
 #   Column      Non-Null Count  Dtype
---  --
 0   Release_Date  25552 non-null    int32
 1   Title         25552 non-null    object
 2   Popularity    25552 non-null    float64
 3   Vote_Count   25552 non-null    int64
 4   Vote_Average  25552 non-null    category
 5   Genre        25552 non-null    category
dtypes: category(2), float64(1), int32(1), int64(1), object(1)
memory usage: 760.4+ MB

In [38]: df.sample()

Out[38]:
Release_Date  199
Title         9415
Popularity    8888
Vote_Count    3005
Vote_Average  4
Genre         19
dtype: int64

Data Visualization

In [39]: sns.set_style('whitegrid')

• What is the most frequent genre in the dataset?

In [40]: df['Genre'].describe()

Out[40]:
count      25552
unique      19
top        Drama
freq       3715
Name: Genre, dtype: object

In [42]: sns.catplot(y = 'Genre', data = df, kind = 'count',
                    order = df['Genre'].value_counts().index, color = '#4287F5')
plt.title('Genre column distribution')
plt.show

Out[42]:
<function matplotlib.pyplot.show(close=None, block=None)>

Genre column distribution

```

- we can notice from the above visual that Drama genre is the most frequent genre in our dataset and has appeared more than 14% of the times among 19 other genres.

- What genres has highest votes ?

```
In [44]: df.head()

Out[44]:
   Release_Date  Title  Popularity  Vote_Count  Vote_Average  Genre
0    2021  Spider-Man: No Way Home    5083.954      8940      popular  Action
1    2021  Spider-Man: No Way Home    5083.954      8940      popular  Adventure
2    2021  Spider-Man: No Way Home    5083.954      8940      popular  Science Fiction
3    2022      The Batman    3827.658      1151      popular  Crime
4    2022      The Batman    3827.658      1151      popular  Mystery

In [47]: sns.catplot(y = 'Vote_Average', data = df, kind = 'count',
                    order = df['Vote_Average'].value_counts().index)
plt.title('Vote_Average')
plt.show()

Vote_Average
average
popular
below_avg
not_popular
count
```

- What movie got the highest popularity ? what's its genre ?

```
In [46]: df.head()

Out[46]:
   Release_Date  Title  Popularity  Vote_Count  Vote_Average  Genre
0    2021  Spider-Man: No Way Home    5083.954      8940      popular  Action
1    2021  Spider-Man: No Way Home    5083.954      8940      popular  Adventure
2    2021  Spider-Man: No Way Home    5083.954      8940      popular  Science Fiction
3    2022      The Batman    3827.658      1151      popular  Crime
4    2022      The Batman    3827.658      1151      popular  Mystery

In [47]: df[df['Popularity'] == df['Popularity'].max()]

Out[47]:
   Release_Date  Title  Popularity  Vote_Count  Vote_Average  Genre
0    2021  Spider-Man: No Way Home    5083.954      8940      popular  Action
1    2021  Spider-Man: No Way Home    5083.954      8940      popular  Adventure
2    2021  Spider-Man: No Way Home    5083.954      8940      popular  Science Fiction

• Q4: What movie got the lowest popularity ? what's its genre?

In [48]: df[df['Popularity'] == df['Popularity'].min()]

Out[48]:
   Release_Date  Title  Popularity  Vote_Count  Vote_Average  Genre
25546    2021  The United States vs. Billie Holiday    13.354       152      average  Music
25547    2021  The United States vs. Billie Holiday    13.354       152      average  Drama
25548    2021  The United States vs. Billie Holiday    13.354       152      average  History
25549    1984      Threads    13.354       186      popular  War
25550    1984      Threads    13.354       186      popular  Drama
25551    1984      Threads    13.354       186      popular  Science Fiction

• Which year has the most filmed movies?

In [49]: df['Release_Date'].hist()
plt.title('Release_Date column distribution')
plt.show()

Release_Date column distribution

```

- Q1: What is the most frequent genre in the dataset?
- Q2: What genres has highest votes ?
- we have 25.5% of our dataset with popular vote (6520 rows). Drama again gets the highest popularity among fans by being having more than 18.5% of movies popularities.
- Q3: What movie got the highest popularity ? what's its genre ?
- Spider-Man: No Way Home has the highest popularity rate in our dataset and it has genres of Action , Adventure and Science Fiction .
- Q4: What movie got the lowest popularity ? what's its genre ?
- The united states, thread" has the highest lowest rate in our dataset and it has genres of music , drama , 'war', 'sci-fi' and history' .
- Q5: Which year has the most filmed movies?
- year 2020 has the highest filming rate in our dataset.